Null Object

Safe logger when monitoring disabled
Season 4 · Enterprise · Intermediate

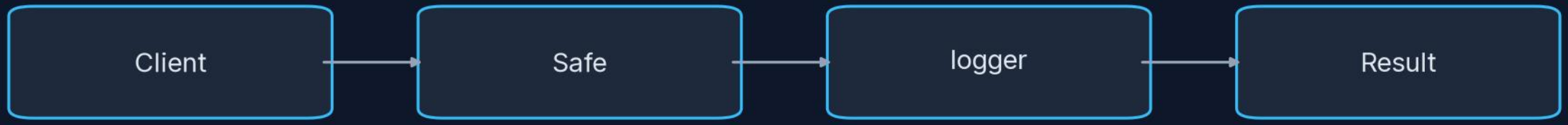
Today's Hook

- Problem: Safe logger when monitoring disabled
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Null Object with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Safe logger when monitoring disabled
- Constraints: solve «Safe logger when monitoring disabled» with Null Object — keep the demo runnable and the failure path observable.
- Naive pain: Naive: logger may be null — every call site null-checks.
- Success looks like: [placed o1]

Structure



Null Object: Safe logger when monitoring disabled

Real-World Scenario

- Running example: Safe logger when monitoring disabled
- Key types: Logger, ConsoleLogger, NullLogger, OrderService
- Happy-path stdout cue: [placed o1]
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.enterprise.nullobject.problem;

/** Naive: logger may be null — every call site null-checks. */
public final class NaiveOrderService {
  private final Logger log;
  public NaiveOrderService(Logger log){ this.log=log; }
  public void place(String id){ if(log!=null) log.info("placed "+id); }
}
```

Why Naive Hurts

- Naive: logger may be null — every call site null-checks.
- Without Null Object, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.enterprise.nullobject.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.concurrent.locks.*;
import java.util.function.*;

interface Logger { void info(String m); }
final class ConsoleLogger implements Logger { private final List<String> lines=new ArrayList<>();
  public void info(String m){ lines.add(m); } List<String> lines(){ return List.copyOf(lines); } }
final class NullLogger implements Logger { public void info(String m){ /* no-op */ } }
final class OrderService {
  private final Logger log; OrderService(Logger log){ this.log=Objects.requireNonNull(log); }
  void place(String id){ log.info("placed "+id); }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.enterprise.nullobject.problem;
import java.util.*; import java.util.concurrent.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Null Object: safe logger ===");
    System.out.println("--- naive null check ---");
new NaiveOrderService(null).place("o1");
System.out.println("--- null object ---");
var console=new ConsoleLogger();
new OrderService(console).place("o1");
new OrderService(new NullLogger()).place("o2");
System.out.println(console.lines());
  }
}
```

Walkthrough Output

```
=== Null Object: safe logger ===
--- naive null check ---
--- null object ---
[placed o1]
```

Step-by-Step Trace

- Match stdout to stages — look for: [placed o1]
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp … 28-null-object ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Null Object
// Prefer records + sealed where they shrink the demo.
// Problem: Safe logger when monitoring disabled
```

Trade-offs

- Use Null Object when «Safe logger when monitoring disabled» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Null Object in product language.

Related Patterns

- Name the pattern usually confused with Null Object and separate the forces.
- Sketch the roles for «Safe logger when monitoring disabled» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Safe logger when monitoring disabled
- Naive pain → Null Object boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.